Bibliography

Recovering the Hidden State of a Stochastic Double-Well System Using Diffusion Maps

---

Primary Source

Banisch, R., Trstanova, Z., Bittracher, A., Klus, S., & Koltai, P. (2017). Diffusion maps tailored to arbitrary non-degenerate Itô processes. *arXiv preprint* arXiv:1710.03484 [math.DS]. (Later published in *Applied and Computational Harmonic Analysis*, 48(1), 242–265, 2020.)

Generalizes classical diffusion maps to approximate the generators of arbitrary non-degenerate Itô diffusions while removing sampling-density bias; motivates recovering the slow reaction coordinate of an SDE such as the double well studied here.

Additional Sources

Coifman, R. R., & Lafon, S. (2006). Diffusion maps. *Applied and Computational Harmonic Analysis*, 21(1), 5–30.

The foundational diffusion-maps paper: introduces the anisotropic (α) kernel-normalization family and the diffusion-distance embedding implemented from scratch in this project.

Belkin, M., & Niyogi, P. (2003). Laplacian eigenmaps for dimensionality reduction and data representation. *Neural Computation*, 15(6), 1373–1396.

Establishes the graph-Laplacian / spectral-embedding framework on which diffusion maps build.

Nadler, B., Lafon, S., Coifman, R. R., & Kevrekidis, I. G. (2006). Diffusion maps, spectral clustering and reaction coordinates of dynamical systems. *Applied and Computational Harmonic Analysis*, 21(1), 113–127.

Connects the leading diffusion coordinates to the slow reaction coordinates and metastable states of stochastic dynamical systems — the basis for expecting the first diffusion coordinate to recover the double-well 'which-well' variable.

Higham, D. J. (2001). An algorithmic introduction to numerical simulation of stochastic differential equations. *SIAM Review*, 43(3), 525–546.

An accessible, widely cited introduction to the Euler–Maruyama scheme used to simulate the double-well SDE $dX = (X - X^3)\,dt + \sigma\,dW$.